Java Programming

Practice Programs

Name: A Sybil Litzy

Reg no:192421110

1. Matrix Multiplication:

Q1 **Matrix multiplication**

Matrix multiplication

Your Code

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);
        int r1 = sc.nextInt();
        int c1 = sc.nextInt();

        int r2 = sc.nextInt();
        int c2 = sc.nextInt();

        if (c1 != r2) {
            System.out.println("Matrix multiplication not possible");
            return;
        }

        int[][] A = new int[r1][c1];
```

```java

        int[][] A = new int[r1][c1];
        int[][] B = new int[r2][c2];
        int[][] C = new int[r1][c2];


        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c1; j++) {
                A[i][j] = sc.nextInt();
            }
        }

        for (int i = 0; i < r2; i++) {
            for (int j = 0; j < c2; j++) {
                B[i][j] = sc.nextInt();
            }
        }
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                for (int k = 0; k < c1; k++) {
                    C[i][j] += A[i][k] * B[k][j];
                }
            }
        }
        for (int i = 0; i < r1; i++) {
```

```java
            for (int j = 0; j < c2; j++) {
                B[i][j] = sc.nextInt();
            }
        }
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                for (int k = 0; k < c1; k++) {
                    C[i][j] += A[i][k] * B[k][j];
                }
            }
        }
        for (int i = 0; i < r1; i++) {
            for (int j = 0; j < c2; j++) {
                System.out.print(C[i][j] + " ");
            }
            System.out.println();
        }

        sc.close();
    }
}
```

Input

```
2 2
2 2
1 2
3 4
```

Output

```
19 22
43 50
```

2. Largest Numbers :

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[n];

        for (int i = 0; i < n; i++)
            a[i] = sc.nextInt();

        int max = a[0];

        for (int i = 1; i < n; i++) {
            if (a[i] > max)
                max = a[i];
        }

        System.out.println("Largest element = " + max);
    }
}
```

Input

```
5
10 25 8 40 15
```

Output

```
Largest element = 40
```

3. Smallest Numbers :

```java
import java.util.Scanner;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[n];

        for (int i = 0; i < n; i++)
            a[i] = sc.nextInt();

        int min = a[0];

        for (int i = 1; i < n; i++) {
            if (a[i] < min)
                min = a[i];
        }

        System.out.println("Smallest element = " + min);
    }
}
```

Input

```
5
10 20 6 35 50
```

Output

```
Smallest element = 6
```

4. NTH Largest Number :

```
import java.util.Scanner;
import java.util.Arrays;

public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[n];

        for (int i = 0; i < n; i++)
            a[i] = sc.nextInt();

        int k = sc.nextInt();

        Arrays.sort(a);

        System.out.println(a[n - k]);
    }
}
```

Input

```
5
10 25 8 40 15
2
```

Output

```
25
```

5. Plindrome number/String:

```java
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[20];
        int count = 0, temp = n;

        while (temp > 0) {
            a[count++] = temp % 10;
            temp /= 10;
        }

        boolean palindrome = true;
        for (int i = 0; i < count / 2; i++) {
            if (a[i] != a[count - 1 - i]) {
                palindrome = false;
                break;
            }
        }

        if (palindrome)
            System.out.println("Palindrome");
        else
            System.out.println("Not Palindrome");
```

Input

```
121
```

Output

```
Palindrome
```

6. Removing Duplicates in array:

```java
import java.util.Scanner;
public class Main {
    public static void main(String[] args) {
        Scanner sc = new Scanner(System.in);

        int n = sc.nextInt();
        int[] a = new int[n];

        for (int i = 0; i < n; i++)
            a[i] = sc.nextInt();

        for (int i = 0; i < n; i++) {
            boolean duplicate = false;

            for (int j = 0; j < i; j++) {
                if (a[i] == a[j]) {
                    duplicate = true;
                    break;
                }
            }

            if (!duplicate)
                System.out.print(a[i] + " ");
        }
```

Input

```
7
10 20 10 30  20 40 50
```

Output

```
10 20 30 40 50
```